

Dhanush S J

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SUMMARY

Software Engineering student with hands-on experience in prompt engineering, LLM integration, and agentic AI pipelines. Proficient in designing and refining reusable prompt templates using zero-shot, few-shot, chain-of-thought, and role-based techniques across Gemini 1.5 Flash, ChatGPT, and Claude. Built real-world AI projects involving multimodal Generative AI, RAG, AI Agents, LangChain, and automation workflows using Python, Flask, Node.js, and REST APIs. Passionate about Generative AI and leveraging prompt engineering to build intelligent, automated systems.

EDUCATION

Vellore Institute of Technology M.Tech Integrated Software Engineering	Expected 2027 CGPA: 8.13
NKM Higher Secondary School Class XII	2022 77%
Desia Matric High School Class X	2020 93.8%

TECHNICAL SKILLS

Prompt Engineering: Zero-shot, Few-shot, Chain-of-thought, Role-based Prompting, Structured Output Generation, Reusable Prompt Libraries, Output Evaluation & Refinement

Generative AI & LLMs: Gemini 1.5/2.5 Flash, ChatGPT, Claude, Cursor, GitHub Copilot, LLM Integration, Agentic AI, Multimodal AI, RAG, AI Agents, LangChain, MCP, OpenAI Codex API

Programming: Python, JavaScript (Node.js), Java, SQL

Backend & APIs: Node.js (Express), Flask, REST APIs, JSON, API Testing (Postman), Twilio API, Cloudinary API

Databases: MySQL, MongoDB, Firebase, NoSQL

Containerization & DevOps: Docker, Kubernetes, CI/CD Pipelines, Git, GitHub, Linux

Cloud Platforms: GCP (Vertex AI), AWS, Azure, Firebase

ML & Deep Learning: TensorFlow, PyTorch, Scikit-Learn, CNN, Random Forest, Decision Tree, Transfer Learning

Software Engineering: SDLC, Agile/Scrum, OOP, Data Structures & Algorithms, System Design

PROJECTS

- AI Resume Analyzer and ATS Job Matcher** | *Python, Flask, Gemini 2.5 Flash, PyPDF2, Tailwind CSS*
- Built an AI-powered resume evaluation platform using Gemini 2.5 Flash; formulated structured prompt templates for ATS scoring, resume-JD match analysis, skill-gap detection, and personalized career recommendations.
 - Created a reusable prompt library covering interview readiness assessment, matching skill extraction, and improvement suggestions; iteratively evaluated and refined LLM outputs for consistency, accuracy, and relevance.
 - Launched a Flask-based web application with PDF resume uploads via PyPDF2, real-time AI feedback generation, and configurable LLM generation parameters for prompt optimization.
- AI Voice Assistant with Multilingual NLG/NLU Support** | *Python, Gemini 1.5 Flash, SpeechRecognition, pyttsx3*
- Built reusable prompt templates and intent-routing logic using zero-shot and role-based prompting to dispatch voice commands to Gemini 1.5 Flash for accurate NLG responses across multiple languages.
 - Optimized LLM prompts for multilingual text generation; systematically evaluated AI-generated outputs for quality, accuracy, and relevance, iterating on prompt design to improve response consistency.
 - Integrated Google STT and pyttsx3 TTS with ambient noise adjustment and turn-detection logic; applied chain-of-thought prompting for complex multi-step voice queries.
- Voice-Controlled Multimodal Vision System** | *Python, Flask, Gemini Vision, PyTorch, SpeechRecognition*
- Constructed a multimodal AI pipeline using structured prompt engineering to route NLU voice commands to Gemini Vision API; crafted few-shot prompts to improve scene description quality and output accuracy.
 - Implemented intent classification to map commands to prompt-driven functions or fall back to LLM for open-ended queries; configured ambient noise adjustment and retry logic for robust real-world performance.
- Event-Driven IoT Surveillance and AI Threat Analysis Pipeline** | *Python, ESP32-CAM, Gemini AI, Firebase, Telegram*
- Crafted Gemini AI prompts to analyze captured footage and generate structured natural language threat summaries; orchestrated real-time Telegram alert and Firebase logging workflows on anomaly detection events.
 - Architected a motion-triggered pipeline keeping the system dormant between events; structured a layered backend separating device interaction, AI inference, and data persistence.
- AI-Based Image Processing and Cloud Storage Platform** | *Python, Flask, Gemini 1.5 Flash, Cloudinary, Firebase*
- Integrated Gemini 1.5 Flash Vision API with optimized prompts for image caption generation; iteratively refined prompt design to improve description quality and evaluate output accuracy.
 - Built REST API endpoints handling image upload, Cloudinary transformation, and Firebase metadata storage.
- Real-Time UPI Fraud Detection and Payment Containment System** | *Node.js, Express, MongoDB, Razorpay*
- Built a pre-settlement risk scoring engine using velocity detection, sender/receiver trust profiling, and blacklisting; implemented a policy engine mapping risk scores (0–100) to Complete / Hold / Reverse decisions.
 - Introduced a soft-hold mechanism for medium-risk transactions with auto-reversal on timeout; delivered an admin analytics dashboard with real-time fraud tracking and structured audit logs in MongoDB.

CERTIFICATIONS & ACHIEVEMENTS

- MongoDB Associate Developer Certification** — In Progress (*MongoDB University*); covering CRUD, Aggregation, and Indexing.
- LeetCode** — Solved 100+ DSA problems covering Arrays, HashMaps, Trees, Recursion, Sliding Window, and Searching.